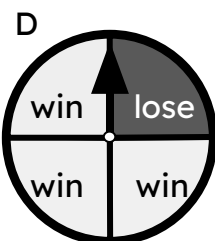
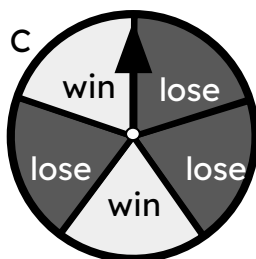
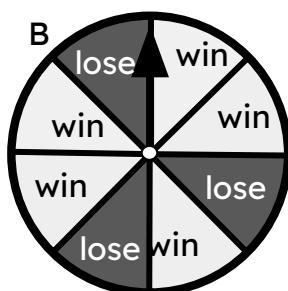
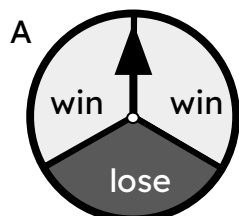




Calculating theoretical probabilities from lists (one event)

Task A: Finding probabilities of events with spinners

1) Here are four win/lose spinners.



a) For each spinner, find the probability that it lands on 'win' in a single spin.

A: $\frac{2}{3}$ B: $\frac{5}{8}$ C: $\frac{2}{5}$ D: $\frac{3}{4}$

Other equivalent fractions, decimals or percentages may be used.

b) Sofia wants to choose the spinner that gives her the best chance of winning. Which spinner should she choose?

spinner D

c) Which spinner is least likely to land on win?

spinner C

2) Alex, Jun and Sam spin the spinner 40 times in total.

They get a point every time the spinner lands on their name.

a) Who should expect to get the most points? *Sam*

b) Who should expect to get the fewest points? *Alex*

c) What is the probability the spinner lands on Alex?

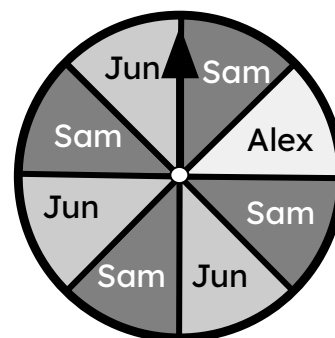
$\frac{1}{8}$

d) What is the probability the spinner lands on Jun?

$\frac{3}{8}$

e) What is the probability the spinner lands on Sam?

$\frac{4}{8}$



Other equivalent fractions, decimals or percentages may be used.

f) How many more points should Sam expect to get compared to Alex?

$\frac{4}{8}$ of 40 = 20

$\frac{1}{8}$ of 40 = 5

$20 - 5 = 15$

g) Will your answer to (f) necessarily happen when they spin the spinner? Explain your answer.

No. These results are only theoretical. The actual results may differ due to chance.



Task B: Finding probabilities of events

- 1) Andeep and Lucas each have a deck of cards spelling out their own name. They play a game where they each randomly select a card from their deck, record the letter and then return the card to the deck.



They win a point if they draw a card containing the letter A.

- a) What are the probabilities for each player winning a point?

Andeep: $\frac{1}{6}$

Lucas: $\frac{1}{5}$

- b) Who is most likely to get a point when they select a card? *Lucas*

- c) They change the rules so that they get a point if they select a vowel.

Who is most likely to get a point with these rules? Explain your answer.

Andeep

The probability that Andeep picks a vowel is $\frac{3}{6}$ while the probability that Lucas picks a vowel is $\frac{2}{5}$ and $\frac{3}{6} > \frac{2}{5}$.

or The probability that Andeep picks a vowel is a half, while the probability that Lucas picks a vowel is less than a half.

- 2) A regular six-sided dice is rolled. Find the following probabilities.

a) $P(3) = \frac{1}{6}$

b) $P(4) = \frac{1}{6}$

c) $P(\text{not } 4) = \frac{5}{6}$

d) $P(\text{even}) = \frac{3}{6}$

e) $P(\text{odd}) = \frac{3}{6}$

f) $P(\text{multiple of } 3) = \frac{2}{6}$

g) $P(7) = 0$

h) $P(\text{not } 7) = 1$

i) $P(\text{factor of } 60) = 1$

Other equivalent fractions, decimals or percentages may be used.

- j) The dice is rolled 60 times. Approximately how many times could it be expected to land on the number 3, based on its probability?

$\frac{1}{6}$ of 60 = 10

- k) Will your answer to part (j) necessarily happen? Explain your answer.

No. The probability only provides a theoretical expectation.

Examples from the simulation:

